

# Economics of Information

Marie-Louise Vierø

Microeconomics 2

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- Welcome to the second half of the course (the contract theory part). Knowledge from the first half is necessary for the second half.

Important learning strategies:

- Ask questions and work with the material!
- Try to solve exercises on your own BEFORE class (TØ).
- Answer all questions and exercises on the slides.
- Try to replicate the calculations from class.
- Ask questions, use instructors, be active in learning!

# Practicalities II

Learning outcome and competences: After completing this course, the students should be able to:

- Translate economic problems into formal game forms and analyze them.
- **Identify and reflect upon the information structure of a problem.**
- Reflect upon the game theoretic equilibrium concepts and their relevance for a given problem.
- **Identify and reflect upon different types of asymmetric information.**
- **Outline the main problems caused by asymmetric information and possible ways to alleviate them.**
- **Analyze economic models of contracts and reflect upon the underlying assumptions.**

- Readings:
  - “An Introduction to the Economics of Information: Incentives and Contracts”  
by: Ines Macho-Stadler, J. David Perez-Castrillo
  - + slides + your notes taken in class
  - READ!
- Agenda:
  - Risk attitude
  - Principal-Agent problems
  - Moral Hazard
  - Adverse Selection
  - Signalling

# von Neumann Morgenstern utility functions and risk preferences

The decision maker (DM) has preferences over probability distributions, or lotteries, over prizes.

- Prizes:  $x_1, x_2, \dots, x_n \in X$
- Lottery:  $p = (p_1, p_2, \dots, p_n)$ ,  $p_i \geq 0$  for all  $i$  and  $\sum_{i=1}^n p_i = 1$ .  $p_i$  is the probability of  $x_i$ .
- von Neumann Morgenstern (vNM) utility function:  
$$V(p) = \sum_{x_i \in \text{supp } p} p(x_i) u(x_i).$$

The vNM utility function represents preferences:

$$\forall p, q, p \succsim q \Leftrightarrow V(p) \geq V(q).$$

# von Neumann Morgenstern utility functions

- If DM is indifferent between  $p$  and  $q$ , then DM is also indifferent to any mixture of the two.
- So indifference curves are linear in probabilities.
- Note: E.g.  $(V(p))^2$  does not represent the same preferences.
- Permit transformations of the form  $aV(p) + b$  for any  $a > 0$  and  $b \in \mathbb{R}$ .
- We have cardinal properties of the utility function.
- Because the curvature measures the DM's risk preferences.

# Risk preferences

Now think about prizes in dollars. So prizes are wealth levels.  
Then  $X = \mathbb{R}_+$ .

- Let  $\delta_x$  denote the lottery that pays  $x$  with probability 1.
- Assume  $x > y \Rightarrow \delta_x \succ \delta_y$  for all  $x, y \in \mathbb{R}_+$ .
- So  $u(x)$  is strictly increasing.
- Let  $E_p = \sum_{x_i \in \text{supp } p} x_i p(x_i)$ .
- Let  $\Delta(X)$  be the set of lotteries over  $X$ .

## Proposition 1

*For all  $p \in \Delta(X)$ ,  $\delta_{E_p} \succsim p$  if and only if the function  $u : X \rightarrow \mathbb{R}$  representing these preferences is concave.*

Which level of wealth would give indifference with the lottery  $p$ ?

## Definition 1

The **certainty equivalent**  $c(p)$  for  $p \in \Delta(X)$ , is  $x \in X$  such that  $\delta_x \sim p$ .

How much money would the DM would pay to avoid the gamble?

## Definition 2

The **risk premium**  $R(p)$  for  $p \in \Delta(X)$ , is  $E_p - c(p)$ .

## Definition 3

The **coefficient of absolute risk aversion** is  $\lambda(w) = -\frac{u''(w)}{u'(w)}$ .

- It is a measure of risk aversion.
- It is independent of which linear transformation of the utility function we take:  $\lambda$  depends on preferences, not on the transformation.

**Important:** The cardinal property risk aversion can be compared over individuals, but we still cannot compare utility levels.

# What is contract theory? - Conflicts of interest

In this lecture, we start with contract theory (or “incentive theory” or “principal-agent theory”). In contract theory we analyse how individuals can organize the exchange of goods and services in the presence of **asymmetric information**. (E.g. work and wage, insurance and money, etc.)

**Many economic situations are characterized by informational asymmetries between economic actors.**

⇒ If there is some **conflict of interest** between the actors, such informational asymmetries can lead to incentive problems (because actors maximize their own utility).

We study such informational asymmetries between **two parties**:

**Principal** ( $P$ ): typically the uninformed party

**Agent** ( $A$ ): informed party

*“The starting point of incentive theory (contract theory) corresponds to the problem of delegating a task to an agent with private information.*

*This private information can be of two types:*

- either the agent can take an action unobserved by the principal, **the case of moral hazard or hidden action**;*
- or the agent has some private knowledge about his cost or valuation that is unobserved by the principal, **the case of adverse selection or hidden knowledge**.*

*Incentive theory considers when this private information is a problem for the principal, and what is the optimal way to cope with it.”<sup>1</sup>*

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<sup>1</sup>Taken from Laffont, Jean-Jacques, and David Martimort (2002): The Theory of Incentives, Princeton University Press.

We consider the **consequences of asymmetric information** about essential economic characteristics of the agent

- Many situations are characterized by asymmetric information:

- **Principal**

- 1 Employer
- 2 Creditor
- 3 Insurance Company
- 4 Government
- 5 People

- **Agent**

- 1 Employee
- 2 Borrower
- 3 Insurance Holder
- 4 People
- 5 Government

- **2 cases** of informational asymmetries:
  - Hidden action
  - Hidden information

# Asymmetric information and efficiency

We are interested in the consequences of asymmetric information.  
Does it change an efficient market to an inefficient market?

- What do we mean by efficiency?
- One of the implicit assumptions of the welfare theorems is that the **characteristics of all commodities are observable** to all market participants.

**Without this condition, distinct markets cannot exist for goods having distinct characteristics, and so the complete markets assumption cannot hold.**

- Without complete markets, the conditions for trade are challenged. Essentially there is uncertainty about which product you are buying.
- Welfare Theorems: First: any CE is efficient  
Second: any efficient allocation can be achieved through CE

*With asymmetric information the market may not reach the efficient outcome.*

Questions we will look at in relation to this are:

- What is asymmetric information about?
- When does it arise?
- Consequences of asymmetric information on efficiency of an allocation?
- How to design **incentive schemes** that influence the agent's incentives in the desired way?
- What does the optimal incentive scheme look like?
- Constrained Pareto optimal allocations: Allocations that cannot be Pareto improved upon by a central authority who cannot observe the individuals' private information.

## Sidestep: History of thought

- Efficient exchange contracts without uncertainty (general equilibrium).
- Efficient exchange contracts with uncertainty (1950s).
- Contractual situations with asymmetric information (1970s and 1980s).
- Dynamic contracting, renegotiation-proofness (1980s and 1990s).
- Incomplete contracts, mechanism design (1990s).
- Behavioural contract theory, empirical and experimental examination of contract theory.

# Hidden information

- Asymmetric information already exists before the parties sign a contract.
- There is asymmetric information about essential economic characteristics of the agent (e.g. valuation of unemployment, health etc.).
- Asymmetric information may lead to a problem of **adverse selection** when the informed party's information affects their market behaviour.
- Involved parties can attempt to circumvent these potential problems.
- We will study two possible contracts that attempt to do so (both dynamic games of incomplete information):
  - **Signalling**: the informed party moves first
  - **Screening**: the uninformed party moves first

# Hidden action

- Symmetric information at the stage of contracting.
- Asymmetry of information arises during the relationship.
- Agent takes an action that cannot be observed by principal.
- Principal observes only a noisy signal of the action. This may lead to problems of moral hazard. (Moral hazard with hidden action.)
- Hybrid case: Same timing as above with hidden information: The principal observes the agent's action, but asymmetric information arises during the relationship about key characteristics in the environment, so that the principal cannot judge whether the action is adequate.

# Contract theory vs Game theory

- While in game theory we were concerned with the equilibrium of a particular game, we now ask how we can structure a game so that it yields us a certain equilibrium outcome.
  - we therefore use many of the tools developed through game theory
- Contract theory is now used in many research areas: labour economics, health economics, management science, corporate finance/governance, and development economics.